

Marvin Durango Cogollo

Digital Twin & Simulation Engineer · VR/XR Developer — Fluid Power, Smart Manufacturing, IoT

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PROFILE

I build the full Digital Twin stack and make it pay off: physics models at the right fidelity (CFD down to real-time), estimation that replaces sensors, industrial connectivity and edge computing, robotics and discrete-event simulation (Visual Components, NVIDIA Isaac Sim), and VR/XR interfaces that let people see and operate the machine, written in Python and C++ — with the safety logic and documentation a team needs to rely on it. I prefer the simplest model that answers the question, and I prove it against measured data.

KEY RESULTS

- Raised capacity predictability by **89%** and cut operator fatigue by **over 50%** with collaborative-robot workflows at Hitachi.
- Delivered a live, remotely operable hydraulic Digital Twin: **<7% tracking error, 186 μ s** real-time estimation on a Raspberry Pi, operated from HoloLens 2.
- Cut cost across the stack: zero-capital pump automation, open-source model replacing MATLAB (<1.3% error), meshless CFD without remeshing.

EXPERIENCE

Digital Twin Research Engineer (Graduate Researcher, Ph.D.)

Purdue University, School of Engineering Technology

Aug 2021 – Present

West Lafayette, IN

- **Built a hydraulic crane and its Digital Twin from the ground up:** HPU with VFD, proportional valves and drivers, boom cylinder, hydraulic swing motor, EHA-driven load brake, pressure/position/torque/encoder instrumentation, DAQ and edge computer — a complete platform for testing control, estimation and autonomy on real hydraulics.
- **Model the core hydraulic building blocks** — EHAs, hydrostatic transmissions, cylinders, hydraulic motors and proportional valves — as lumped-parameter models in Python identified from bench data, reusable for sizing, control design and Digital Twins.
- **Proved the twin matches the machine:** validated model predictions against the instrumented bench with **<7% tracking error** across multiple operating conditions.
- **Replaced a physical sensor with software:** a real-time Extended Kalman Filter (Python) reconstructs actuator velocity and pressures on a Raspberry Pi 4 in **186 μ s per update** — no velocity sensor, no desktop PC.
- **Turned off-the-shelf proportional valves into calibrated models** from in-place step tests (flow-gain fit $R^2 = 0.997 / 0.973$, direction-specific deadbands) — no disassembly, flow bench or vendor data.
- **Operate a real machine from mixed reality:** on HoloLens 2 the operator drags the crane's tip by hand; inverse kinematics (FABRIK) converts it to joint targets streamed over TCP, and the physical boom and swing follow.
- **Industrial network and automation:** integrated a torque sensor and motor drive on one RS-485 Modbus RTU fieldbus (370 transactions, 0 errors at commissioning) and automated the hydraulic power unit through its VFD at zero capital cost — remote start, software speed control and soft-start.
- **Made the rig safe and reliable to run remotely:** supervisory state machine blocks motion until pressure and pump speed are stable, latches faults and enforces travel limits; ended recurring DAQ-board failures by redesigning grounding (star ground, galvanic isolation); wrote maintainer-level SOPs so others can operate and repair the bench.
- **Virtual commissioning of a production line (in progress):** Digital Twin of the 6-station skateboard line in Purdue's Smart Learning Factory (Visual Components, OPC UA), 2 of 6 stations recreated; virtual UR10e validated against the physical robot; shop-floor data streamed to AWS and over MQTT.

R&D Intern — Smart & Digital Manufacturing

Hitachi America, Ltd. — Industrial Innovation Laboratory

Jun 2023 – Jan 2024

Farmington Hills, MI

- **Raised capacity predictability by 89% and cut operator fatigue by over 50%** by programming UR5 collaborative-robot tasks for hybrid human-robot workflows.
- **De-risked robot commissioning:** built a sensor-integrated Digital Twin of a manufacturing cell so robot programs and layout decisions were tested virtually before touching the line.
- **Live robot state in the twin:** streamed UR5 data through ROS into the cell's Digital Twin, so the virtual cell mirrored the physical robot without manual exports.
- **Discrete-event simulation:** modeled packaging processes and demand-forecast scenarios to evaluate throughput and capacity before investing.

Instructor / Graduate Teaching Assistant

Purdue University, School of Engineering Technology

2021 – Present

West Lafayette, IN

- **Designed and taught a project-based course** (instructor of record, Fall 2025: Mixed Reality Smart Manufacturing) in which students build Digital Twins of a full production line — robotic arms, conveyors, AGVs.

- Instructor / TA for Advanced Fluid Power (MET 334) and Pneumatic Control (MET 436): system modeling, instrumentation and control, with simulation and XR to visualize system dynamics in real time.

SELECTED PROJECTS

- **Built a go/no-go guide for SPH in hydraulics** (J. Hydrodynamics, accepted): compared open-boundary methods across 9 studies, pinpointed the missing pressure inlet/outlet capability in open-source tools, and delivered a decision chart for when SPH is worth it vs. mesh CFD.
- **Pneumatic Digital Twin with XR control**: hybrid estimator (EKF + LuGre friction + Bayesian regression) predicts actuator motion from measured signals only; real-time proportional control from a HoloLens; open dataset released.
- **Optimized hydrocyclone separators with coupled CFD-DEM**, grounding the model in particle-size distributions measured by laser diffraction (Malvern Mastersizer) — predictions tied to the actual feed.
- **See what sensors can't**: modeled a gear pump and pressure-relief valve in Ansys CFX and rendered their flow fields in mixed reality, exposing internal velocity and pressure no installed sensor can measure.
- **Built GPU physics from scratch**: implemented XPBD soft-body dynamics in C++ and OpenGL compute shaders — the kind of real-time physics engines such as NVIDIA PhysX provide.

EARLIER EXPERIENCE

- **Visiting Scholar, Purdue (2019–2020)** — designed and built a custom hydraulic test rig; CFD of multiphase reservoir systems. **Graduate Researcher & TA, Universidad de los Andes (2017–2019)** — CFD-DEM hydrocyclone optimization.

SERVICE & LEADERSHIP

- Peer reviewer, MAHA Fluid Power Conference 2026 (RANS turbulence benchmarking; CFD of pressure-regulating valve flow).
- Lead two undergraduate interns (fieldbus retrofit; LiDAR object and distance detection) with scoped plans and deliverables.

TECHNICAL SKILLS

Simulation	Ansys Fluent & CFX, SPH, CFD-DEM, RANS/RSM/LES, FEM, lumped-parameter models, XPBD, robotics and discrete-event simulation, virtual commissioning (Visual Components, NVIDIA Isaac Sim)
Hydraulics	Test-bench design & build; EHAs, hydrostatic transmissions, cylinders, motors, proportional valves; system optimization
Estimation & control	EKF, Bayesian parameter regression, system identification, friction modeling, supervisory state machines, FABRIK IK, RRT*
Robotics	Collaborative robots (UR5, UR10e), robot commissioning, ROS, LiDAR perception (Unitree L1)
IoT & automation	OPC UA, MQTT, Modbus RTU/RS-485, CAN (J1939), TCP/UDP, AWS, Firebase, VFDs, PLCs, Raspberry Pi, Jetson Nano
XR & graphics	Unity 3D (C#), MRTK3, HoloLens 2, GLSL compute shaders, HLSL
ML & optimization	CNNs (transfer learning), YOLO, OpenCV; nonlinear optimization (gradient methods, KKT)
Programming	Python (NumPy, Flask, asyncio), C++, C#, MATLAB/Simulink, JavaScript, Git, Linux
Languages	English, Spanish (bilingual)

EDUCATION

Ph.D., Engineering Technology

Purdue University — dissertation defended Sept 2026

Physics-based Digital Twins for hydraulic machinery. Coursework: nonlinear optimization, machine & deep learning, CFD, FEA.

Dec 2026

West Lafayette, IN

M.S., Mechanical Engineering

Universidad de los Andes — thesis: CFD-DEM optimization of hydrocyclones

2019

Bogotá, Colombia

B.S., Mechanical Engineering

Universidad Tecnológica de Bolívar

2015

Cartagena, Colombia

PUBLICATIONS

- Durango, M., & Garcia-Bravo, J. M. (2026). Review of SPH for Internal Flow in Hydraulic Components. *Journal of Hydrodynamics* (Springer), accepted.
- Durango, M., Garcia, Y., Duncan, S., & Garcia-Bravo, J. M. A Physics-Informed Hybrid Digital Twin Framework with Mixed Reality Interface for Fluid Power Teleoperation: A Pneumatic Proof-of-Concept. *Engineering Science and Technology, an Int. Journal*, under revision.
- Durango, M., & Garcia-Bravo, J. M. (2025). Demonstration of a Digital Twin Framework for a Two-Actuator Hydraulic Application (in press).
- Durango, M., & Garcia-Bravo, J. M. (2024). Physics-Based Digital Twin for Hydraulic Equipment. *DigiTwin Conference*.
- Durango, M., & Garcia-Bravo, J. M. (2023). SPH as a Model for Hydraulic Components. *ASME/BATH Symposium on Fluid Power and Motion Control*.
- Durango, M., Borders, E., Newell, B., Garcia, J., & Breidi, F. (2022). Mixed Reality for Fluid Power Instruction. *ASME Annual Conference*.
- Durango, M., Garcia-Bravo, J., Newell, B., & Gonzalez-Mancera, A. (2020). CFD Modeling of Hydrocyclones. *Fluids*, 5(3), 118.